

Sun-dried raisins are a cost-effective alternative to Sports Jelly Beans in prolonged cycling.



The purpose of this study was to examine the effects of a natural carbohydrate (CHO) source in the form of sun-dried raisins (SDRs) vs. Sports Jelly Beans™ (SJBs) on endurance performance in trained cyclists and triathletes. Ten healthy men (18-33 years) completed 1 water-only acclimatization exercise trial and 2 randomized exercise trials administered in a crossover fashion. Each trial consisted of a 120-minute constant-intensity glycogen depletion period followed by a 10-km time trial (TT). During each experimental trial, participants consumed isocaloric amounts of SDRs or SJBs in 20-minute intervals. Measurements included time to complete 10-km TT, power output during 10-km TT, blood glucose levels and respiratory exchange ratio during glycogen depletion period, rate of perceived exertion (RPE), 'flow' questionnaire responses, and a hedonic (i.e., pleasantness) sensory acceptance test. There were no significant differences in endurance performance for TT time (SDRs vs. SJBs, 17.3 ± 0.4 vs. 17.3 ± 0.4 seconds) or power (229.3 ± 13.0 vs. 232.0 ± 13.6 W), resting blood glucose levels (5.8 ± 0.4 mmol·L⁻¹ for SDRs and 5.4 ± 0.2 mmol·L⁻¹ for SJBs), RPE, or flow experiences between SDR and SJB trials. However, the mean sensory acceptance scores were significantly higher for the SDRs compared to the SJBs (50.7 ± 1.7 vs. 44.3 ± 2.7). Consuming SDRs or SJBs during 120 minutes of intense cycling results in similar subsequent TT performances and are equally effective in maintaining blood glucose levels during exercise. Therefore, SDRs are a natural, pleasant, cost-effective CHO alternative to commercial SJBs that can be used during moderate- to high-intensity endurance exercise.